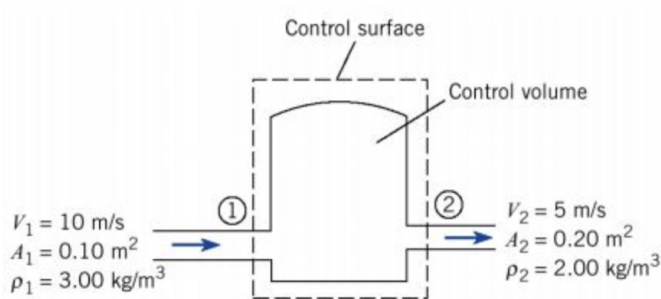

Problem 1. Mass Conservation (10 points)

(a) Explain what “mass conservation” is in your own words. (2 points)

(b) Gas flows into and out of the chamber as shown. What mass flow rate is gas accumulating in (or evacuating from) the chamber? (4 points)

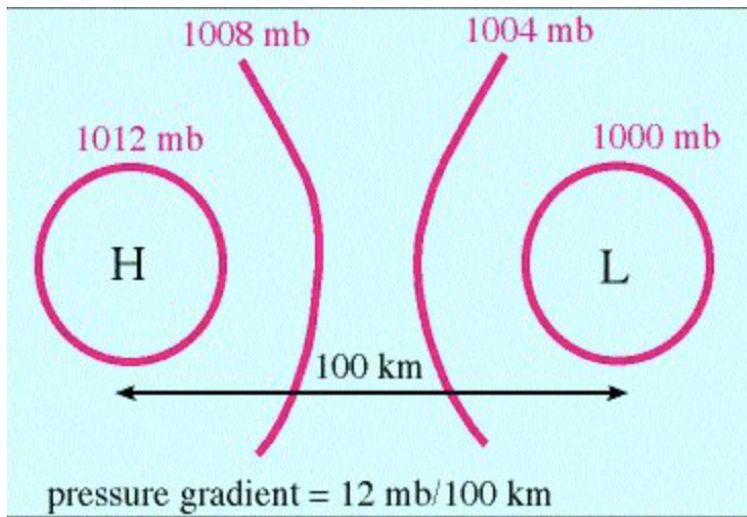


(c) The mass flow rate $\dot{M}$ of a gas flowing in a pipe is affected by the diameter of the pipe “ D ” and the volumetric flow rate $\dot{V}$ of the gas according to the following equations: $\dot{M} = A \times D^2 \times \dot{V}$. Given the unit of $\dot{M}$ as kg/s , D as cm , and $\dot{V}$ as cm^3/s . Find the unit of the constant A (4 points).

Problem 2. Pressure Gradient Force (20 points)

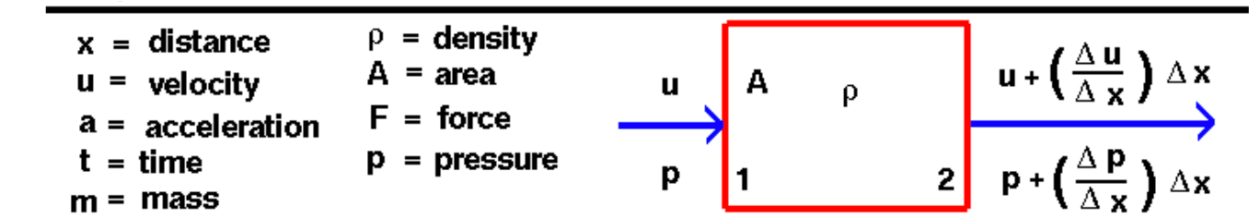
(a) Explain the meaning of “pressure gradient” in your own words. (2 points)

(b) Calculate the pressure gradient shown below and plot the direction of the winds driven by the pressure gradient force. (4 points)



(c) Explain why denser isobaric lines means a greater pressure gradient force (4 points)

(d) Consider an object experiencing only pressure gradient force. Write down the expression of the momentum conservation. Hint: the momentum conservation is $F = m \times a$ and the acceleration $a = \frac{\Delta u}{\Delta t}$. The pressure gradient force has the format of $\frac{1}{\rho} \frac{\Delta p}{\Delta x}$. (8 points)



(e) Explain the meaning of “Ridge” and “Trough” in your own words. (2 points)

Problem 3. *Forces Related to A Rotating Reference Frame (20 points)*

(a) Explain the differences between a non-inertial reference frame and an inertial reference frame. (2 points)

(b) Circle all the correct statements. (3 points)

- (i) The Centripetal force is an apparent force.
- (ii) The Coriolis force is an apparent force.
- (iii) The Centrifugal force is a real force.
- (iv) The Coriolis Effect arises from the fact that Earth rotates.
- (v) In middle latitudes of the Southern Hemisphere, the Coriolis Effect causes a west wind to become a northwest wind.
- (vi) The Coriolis Effect decreases in magnitude with decreasing wind speed and latitude.

(c) A 65 kg runner jogs on a completely circular track with a radius of 60 m. If he completes one lap in 50 seconds, determine the force of friction needed to maintain the circular motion. Hint: the formula for Centripetal force is $F_c = m \frac{v^2}{r}$. (5 points)

(d) The Coriolis force (CF) is defined as $CF = 2\Omega \sin(\phi)v$, where $\Omega = 7.27 \times 10^{-5} s^{-1}$ is the Earth's rotation rate, ϕ denotes the latitude and v (units: m/s) is the speed of motion relative to Earth of the object in this question. Consider a circular bathtub initially filled with water draining through a hole at the bottom center of the bathtub. By unplugging the bathtub we create a low pressure area at the center of the bathtub. Assume the bathtub is located in Atlanta where $\phi = 35^\circ$.

Compute the magnitude of the Coriolis force using equation (1) assuming the water drains with a speed of $v = 5m/s$. Useful information: $\sin(\phi = 35^\circ) = 0.57$. Make sure to include the correct units! (4 points)

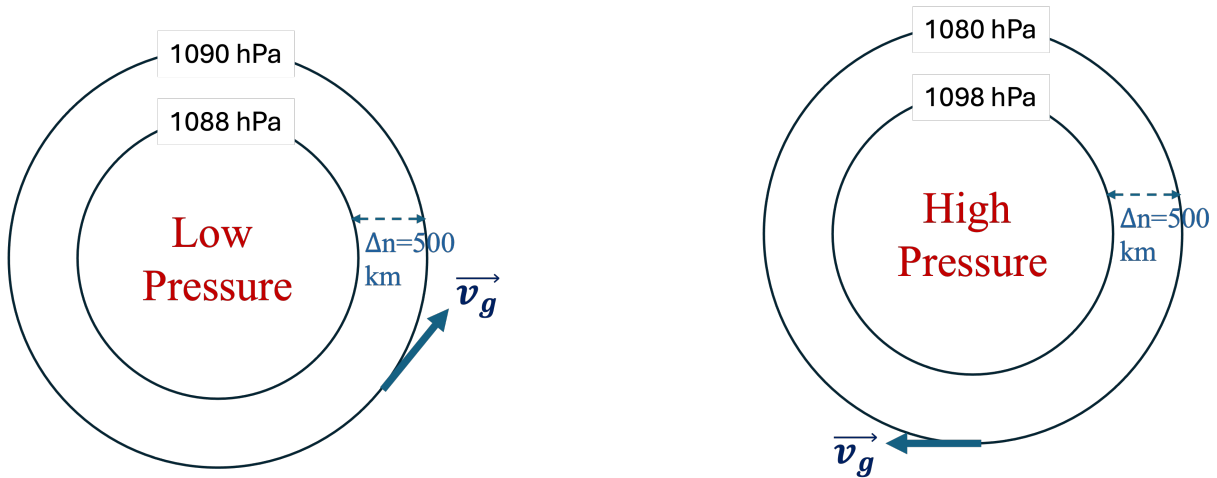
(e) Using your answer to part (d), is the Coriolis force acting on the water either greater than, less than or equal to the force of gravity acting on the water? (2 points)

(f) Now, let's investigate whether the water could have attained a speed of $5m/s$ as a result of the Coriolis force only. The Coriolis force as written in equation (1) has units of acceleration (i.e. m/s^2). So, dividing a speed by CF gives us a time scale. That is, an estimate of the amount of time it would take the Coriolis force to act and make the water move at that speed. Compute the aforementioned time scale using CF found in part (d) and $v = 5m/s$. Report your answer in minutes. (4 points)

Problem 4. Geostrophic Wind Balance and Thermal Wind Balance (20 points)

(a) Explain what determines the geostrophic wind balance (2 points).

(b) Plot the pressure gradient force (PGF) and the Coriolis force (CF) in the following two scenarios: Northern Hemisphere Cyclonic flow and Anticyclonic flow. (4 points)



(c) Calculate the geostrophic wind speed for the Cyclonic flow and the Anticyclonic flow in (b). Assume the perpendicular distance between isobaric lines is $\Delta n = 500 \text{ km}$ in both scenarios and the air density is 1.2 kg/m^3 . Hint: the geostrophic wind speed equation is given by: $|V_g| = \frac{1}{\rho} \cdot \frac{\Delta p}{\Delta n}$. (4 points)

(d) Explain how the cross-isobar flow is formed. (4 points)

(e) Explain why a more tilted isobaric surface means stronger geostrophic winds. (4 points)

(f) In the following scenario, is the geostrophic winds increase or decrease with altitude? Is the wind flowing in or coming out of the page? Explain your reasons. (2 points)

